Quantum Query Algorithms for the Constructive Diagonal Ramsey Theorem

Anonymous author
Anonymous affiliation

Abstract

The constructive diagonal Ramsey problem asks for a clique or an independent set of the size guaranteed by Ramsey's theorem, using adjacency queries to the input graph. Given coherent adjacency-oracle access to an N-vertex simple graph, we give a bounded-error quantum algorithm that, for every $K \geq 2$ and $N \geq 4^{K-1}$, finds and verifies a homogeneous K-set using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries with failure probability at most η. At the Ramsey scale $N = 2^n$, it finds a homogeneous set of order $\lfloor n/2 \rfloor + 1$ using $O(\sqrt{N} \log N \log(\log N/\eta))$ queries. The standard explicit Erdős–Szekeres recursion uses $O(N)$ queries.

The algorithm implements the constructive Ramsey recursion without materializing its nested candidate sets. It represents each candidate set by a short conjunction of adjacency constraints, samples from this implicit set using capped quantum search with an unknown number of marked items, and uses a scale-aware concentration schedule that assigns more accuracy to inexpensive early levels than to expensive late levels. We also give an estimation-free size-biased recursion, extend it to every fixed number of edge colours, and prove a randomized classical lower bound of $\Omega(N^{1-1/\sqrt{2}})$ in the same parameterization. This lower bound does not establish a quantum–classical separation. The result improves the query complexity of finding guaranteed homogeneous sets; it does not improve a numerical Ramsey bound or make a gate-complexity claim.

2012 ACM Subject Classification Theory of computation → Quantum query complexity

Keywords and phrases Diagonal Ramsey theorem, clique or independent set, quantum query complexity, adjacency oracle, implicit sets, total search

1 Introduction

The elementary Erdős–Szekeres proof of the diagonal Ramsey theorem is constructive [7]: choose a pivot, retain a colour neighbourhood, and iterate. At the sufficient threshold $M = 4^{K-1}$, truncating each retained neighbourhood to half the current size gives an $O(M)$-query implementation with explicit candidate sets. This paper shows that the same proof architecture has a substantially cheaper quantum implementation when the nested candidate sets are instead represented by their membership predicates. We use the elementary 4^{K-1} sufficient threshold because it supports this recursion; it is not the best current extremal bound, which has exponential base strictly below four [3].

We work in the promised valid-graph adjacency-oracle model. One coherent query returns the colour of a requested unordered vertex pair. The output is a set of vertices inducing one colour. The main theorem is the following.

▶ **Theorem 1.1** (Implicit-majority quantum Ramsey search). *Let $K \geq 2$, $N \geq 4^{K-1}$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple undirected graph on N vertices, there is a quantum algorithm which, with probability at least $1-\eta$, outputs a K-clique or a K-vertex independent set using*

$$O\Big(2^K K \log \frac{K}{\eta}\Big)$$

edge queries in the worst case. The algorithm verifies every returned witness and may output $\perp$ on the exceptional event.

▶ **Corollary 1.2.** *For a promised simple graph on $N = 2^n$ vertices, one can find a clique or independent set of order $\lfloor n/2 \rfloor + 1$ using*

$$O\Big(\sqrt{N} \log N \log \frac{\log N}{\eta}\Big)$$

edge queries.

The first idea is to use capped unknown-solution quantum search to sample from a set specified only by the $O(K)$ pivot constraints accumulated so far. There are two ways to choose the next colour class. The cleaner, estimation-free method samples one uniform vertex and follows the colour of its edge to the current pivot. This chooses a class with probability proportional to its size. A new survival lemma shows that this size-biased recursion completes with probability greater than $1/2$ from 4^{K-1} vertices, giving an $O(2^K K^2 \log K \log(1/\eta))$-query algorithm.

The sharper method uses a scale-aware concentration budget. Sampling a deep candidate set costs exponentially more than sampling an early one, so we estimate early majorities more accurately and permit larger errors later. The sum of all errors is kept below a fixed constant, whereas the work is balanced across the recursion levels. A dyadic schedule removes two avoidable factors of K from the uniform-accuracy analysis and gives Theorem 1.1. The size-biased argument extends directly to any fixed number of edge colours; we record the multicolour form as Corollary C.1.

Contributions.

The algorithmic contributions are as follows.

- It gives a worst-case $O(2^K K \log(K/\eta))$ quantum-query algorithm for constructive diagonal Ramsey search, or $\widetilde{O}(\sqrt{N})$ at the succinct Ramsey scale.
- It introduces two ways to navigate the implicit recursion: a size-biased rule with a short survival proof, and a sharper scale-aware majority estimator whose accuracy budget is matched to the cost of each level.
- It extends the size-biased mechanism to a fixed number of colours and places the result against a randomized classical lower bound of $\Omega(N^{1-1/\sqrt{2}})$ in the same parameterization.

The result is algorithmic rather than extremal. It does not change any numerical Ramsey bound or construct an avoidance colouring. It gives a square-root query improvement, up to polylogarithmic factors, over the standard $O(N)$ constructive recursion, but no matching lower bound against every randomized classical algorithm is presently known.

For completeness, we address a discrepancy with the closest exact-model source. The FOCS 2024 paper of Jain, Li, Robere, and Xun reports an $N^{1-o(1)}$ quantum-query lower bound for its default Ramsey relation [11, Section III, p. 415]. Under its formal $N = 2^n$, $K = n/2$ convention, this conflicts with Corollary 1.2. Section 6 checks the model alignment and shows that direct substitution into the two cited results does not give the reported exponent. Our algorithm and its proof are independent of that consequence; the comparison does not challenge the paper's principal TFNP separation results.

The rest of the paper defines the oracle model and capped sampler, proves the recurrence and query bound, derives a randomized classical lower bound from the random-Painter theorem, and records the novelty and nonclaim boundaries.

2 Oracle model and capped uniform search

Let $G\colon \binom{[N]}{2} \to \{0,1\}$ be a promised simple undirected graph, extended symmetrically to ordered pairs with $G(u,u) = 0$. The coherent edge oracle acts as

$$O_G|u, v, b\rangle = |u, v, b \oplus G(u, v)\rangle.$$

Only calls to O_G are counted in the query bound. Non-oracle computation, including reversible operations on vertex labels, is not charged in this model. Coherent access to O_G is supplied as part of the input model.

The promised formulation is the clean graph-theoretic core, but the usual locally verifiable TFNP relation also follows at the same asymptotic cost. Given an arbitrary Boolean circuit $C(u, v)$, define the canonical simple graph

$$A(u,v) = \begin{cases} C(\min\{u,v\}, \max\{u,v\}), & u \neq v, \\ 0, & u = v. \end{cases}$$

One A-query uses $O(1)$ C-queries, including uncomputation. Run Theorem 1.1 on A and let W be the returned homogeneous set. Query the $O(K^2)$ diagonal and both ordered entries induced by W. If a self-loop or asymmetry is found, return it as the invalid-encoding certificate. Otherwise C agrees with A on W, so W is a valid clique or independent-set witness. This gives the following model-alignment corollary.

▶ **Corollary 2.1** (Locally verifiable arbitrary encoding). *The standard Ramsey TFNP relation that accepts either a local invalidity certificate or a locally verified homogeneous set has the same $O(2^K K \log(K/\eta))$ quantum-query upper bound.*

We use the following standard consequence of the unknown-solution search of Boyer, Brassard, Hoyer, and Tapp [2]. A detailed formulation is included in Appendix A because bounded termination and conditional uniformity are both needed here.

▶ **Lemma 2.2** (Capped uniform marked-item search). *Let $T \subseteq [M]$ have density at least $\lambda > 0$, and suppose its membership predicate costs q edge queries. There is a quantum block which uses at most $O(q/\sqrt{\lambda})$ edge queries, succeeds with probability at least an absolute constant $p_0 > 0$, and, conditional on verified success, returns an exactly uniform element of T. The same predetermined cap makes the block terminate if the density promise is false or T is empty.*

Moreover, m independent uniform samples can be collected, except with probability δ, using a predetermined $O(m + \log(1/\delta))$ blocks whenever the density promise holds.

The batch statement is what avoids paying a separate $\log(1/\delta)$ factor for each of the many samples. Every block starts from the uniform state, uses fresh internal randomness, measures, and verifies exact membership. Conditioned on its success indicator, the returned identity is uniform; identities from distinct successful blocks are independent.

For comparison, start the exact classical recursion with $M = 4^{K-1} = 2^{r+1}$ vertices, where $r = 2K - 3$. At level i, query all edges from a pivot to the other $s_i = M/2^i$ candidates. A majority colour contains at least $s_i/2$ of the $s_i - 1$ non-pivot vertices; retain exactly $s_i/2$ of them. This explicit truncation ensures $\sum_{i<r}(s_i - 1) < 2M$ queries and leaves two vertices after r rounds. Without truncation, retaining the entire majority class need not give a geometric query sum. The quantum algorithm below retains whole classes but accesses them implicitly.

3 Two implicit-set recursions

3.1 An estimation-free size-biased recursion

The following algorithm isolates the simplest new mechanism. Set $M = 4^{K-1}$ and $r = 2K - 3$, restrict to the first M vertices, let $S_0 = [M]$, and fix any $v_0 \in S_0$. At step $i = 0, \ldots, r-1$, define $T_i = S_i \setminus \{v_i\}$. Use the permutation-symmetric repeated capped search of Lemma 2.2, with the universal density lower bound $1/M$, to sample an exactly uniform $x_i \in T_i$, or abort after a fixed cap. Query

$$c_i = G(v_i, x_i),$$

and set

$$S_{i+1} = \{x \in T_i : G(v_i, x) = c_i\}, \qquad v_{i+1} = x_i.$$

After the last step put $w = v_r$. Among the r labels c_i, choose a colour appearing at least $K - 1$ times and return the corresponding pivots together with w, after direct verification.

The set S_i is never materialized. Its membership predicate consists of the exact edge constraints from the earlier pivots and the exclusions of those pivots. It therefore costs $O(i + 1)$ edge queries. Repeating the capped block $O(\log(2r))$ times makes the conditional failure probability of each nonempty-set sample $O(1/r)$. The structural probability that all r sets remain nonempty is controlled in Lemma 4.1.

3.2 Scale-aware approximate majorities

Fix $M = 4^{K-1}$ input vertices and set

$$r = 2K - 3, \qquad E = \frac{1}{16}.$$

For $0 \le i < r$, define

$$d_i = \left\lceil \frac{r-1-i}{6} \right\rceil, \qquad z_i = 2^{-d_i}, \qquad Z = \sum_{j=0}^{r-1} z_j,$$

and allocate

$$\varepsilon_i = E\frac{z_i}{Z}, \qquad a_i = \frac{1}{2} - \varepsilon_i. \tag{1}$$

Thus $\sum_i \varepsilon_i = E$. The error allowance doubles once every six levels as the search sets become more expensive.

Before round i, retain classical pivots $v_0, \ldots, v_{i-1}$ and colours $c_0, \ldots, c_{i-1}$. They define the exact implicit set

$$S_i = \{u \in [M] \setminus \{v_0, \ldots, v_{i-1}\} : G(v_j, u) = c_j \text{ for every } j < i\}. \tag{2}$$

Its reversible membership predicate uses $O(i)$ edge queries.

At round i, use Lemma 2.2 to obtain a verified pivot $v_i \in S_i$. Then collect, with replacement,

$$m_i = \left\lceil \frac{1}{2\varepsilon_i^2} \log \frac{4r}{\eta} \right\rceil \tag{3}$$

independent uniform samples from $S_i \setminus \{v_i\}$. Query their edges to v_i and write $\widehat{p}_i$ for the observed fraction of colour 1. Set $c_i = 1$ if $\widehat{p}_i \ge 1/2$ and $c_i = 0$ otherwise, and append this exact constraint to Equation (2) to obtain S_{i+1}.

After r rounds, use one final capped search to obtain $w \in S_r$. Select a colour occurring at least $K-1$ times among $c_0, \ldots, c_{r-1}$, take any $K-1$ pivots carrying it, and add w. Before returning, query all $\binom{K}{2}$ induced edges and accept only an exactly homogeneous witness.

Every search uses a predetermined density bound proved in the next section. If an earlier random estimate was bad and a later set violates that density promise, the fixed cap still terminates. The algorithm returns $\perp$ if a search or batch fails to supply its required verified samples; it need not detect a density violation itself. There is no repeat-until-success loop that can hang on an empty set.

4 Correctness and survival of the nested sets

We begin with the estimation-free algorithm. For integers $d, s \ge 0$, let $P_d(s)$ be the infimum, over every graph, every candidate set of size s, and every distinguished pivot in it, of the probability that the ideal size-biased recursion completes d further steps. Put $P_0(s) = 1$ for $s \ge 1$, $P_d(0) = 0$, and $P_d(1) = 0$ for $d \ge 1$.

▶ **Lemma 4.1** (Size-biased survival). *For every $d \geq 0$ and $s \geq 1$,*

$$P_d(s) \geq \frac{(s - 2^d + 1)_+}{s}, \qquad (y)_+ = \max\{y, 0\}.$$

In particular, for $M = 4^{K-1}$ and $r = 2K - 3$, the ideal recursion completes with probability at least $1/2 + 1/M$.

Proof. The case $d = 0$ is immediate. For $d \geq 1$ and $s \geq 2$, let the two colour classes of the $s - 1$ non-pivot vertices have sizes a and b. A uniform sample selects them with probabilities $a/(s-1)$ and $b/(s-1)$, so induction gives

$$P_d(s) \geq \frac{aP_{d-1}(a) + bP_{d-1}(b)}{s-1} \geq \frac{(a - 2^{d-1} + 1)_+ + (b - 2^{d-1} + 1)_+}{s-1}.$$

Since $(x)_+ + (y)_+ \geq (x+y)_+$ and $a+b = s-1$, the numerator is at least $(s - 2^d + 1)_+$. Replacing the denominator $s - 1$ by s only weakens the bound. The definitions cover $s \leq 1$. Finally $M = 2^{r+1}$, so the displayed bound is $(M - 2^r + 1)/M = 1/2 + 1/M$. ◀

Exact conditional uniformity of each successful capped search lets us couple an implemented run to this ideal recursion. Give each of its r searches failure probability at most $1/(100r)$. A union bound reduces the completion probability by at most $1/100$, so one fixed-cap run succeeds with probability at least 0.49. Independent repetition and final exact verification amplify this to $1 - \eta$ using $O(\log(1/\eta))$ runs. On every completed run, nesting gives $G(v_i, v_j) = G(v_i, w) = c_i$ for $j > i$; the pigeonhole argument used in Lemma 4.4 therefore proves exact homogeneity.

We now prove the sharper scale-aware algorithm.

We first isolate the adaptive concentration event.

▶ **Lemma 4.2** (Conditional near-majorities). *Except with probability at most $\eta/2$, every completed sampling batch reached after successful searches and an accurate prefix satisfies*

$$|\widehat{p}_i - p_i| \leq \varepsilon_i,$$

where p_i is the true colour-1 fraction from v_i to $S_i \setminus \{v_i\}$.

Proof. Condition on the complete classical history $\mathcal{H}_i$ before level i's samples. The graph, S_i, and v_i are then fixed. Let $\mathcal{B}_i$ denote verified batch success. Conditional on $\mathcal{B}_i$, Lemma 2.2 gives independent uniform samples from the fixed set. Hoeffding's inequality and Equation (3) give

$$\Pr(|\widehat{p}_i - p_i| > \varepsilon_i \mid \mathcal{H}_i, \mathcal{B}_i) \leq 2e^{-2m_i\varepsilon_i^2} \leq \frac{\eta}{2r}.$$

Multiplying by $\Pr(\mathcal{B}_i \mid \mathcal{H}_i) \leq 1$ gives the same bound for a completed but inaccurate batch conditional on $\mathcal{H}_i$. Apply it to the first inaccurate completed level following an accurate prefix. The tower property removes the conditioning, and a union bound over the r possible levels proves the claim. No independence between levels is used. ◀

There are r pivot searches, r sampling batches, and one final search. Allocate failure probability at most $\eta/(4r+2)$ to each one by the capped amplification in Lemma 2.2. Apply these conditional bounds only after a good prefix, just as in the preceding proof. Their union has probability at most $\eta/2$. Together with Lemma 4.2, every search and estimate needed below is therefore good except with probability at most η.

Let $s_i = |S_i|$ and

$$A_0 = 1, \qquad A_i = \prod_{j=0}^{i-1} a_j.$$

▶ **Lemma 4.3** (Variable recurrence and density)**.** *On every prefix satisfying Lemma 4.2,*

$$s_i > MA_i - 1, \qquad A_i \geq \frac{7}{8} 2^{-i}.$$

Consequently S_r is nonempty, and for every sampling level $i < r$,

$$\frac{|S_i \setminus \{v_i\}|}{M} > \frac{3}{8} 2^{-i}.$$

The final set also satisfies $|S_r|/M \geq 2^{-(r+1)} \geq (3/8)2^{-r}$.

Proof. An accurate empirical majority retains at least fraction $a_i = 1/2 - \varepsilon_i$ of the non-pivot vertices, so

$$s_{i+1} \geq a_i(s_i - 1). \tag{4}$$

Since $\sum_i \varepsilon_i = 1/16$ and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$,

$$A_i = 2^{-i} \prod_{j<i} (1 - 2\varepsilon_j) \geq 2^{-i} \left(1 - 2 \sum_{j<i} \varepsilon_j \right) \geq \frac{7}{8} 2^{-i}.$$

We prove $s_i > MA_i - 1$ by induction. It holds at $i = 0$. If it holds at i, then Equation (4) and $a_i < 1/2$ give

$$s_{i+1} > a_i(MA_i - 2) = MA_{i+1} - 2a_i > MA_{i+1} - 1.$$

Because $M2^{-r} = 2$, the final lower bound is $s_r > (7/8)2 - 1 = 3/4$, and the integer s_r is positive. If $i < r$, then $M2^{-i} \geq 4$ and

$$s_i - 1 > MA_i - 2 \geq \frac{7}{8} M2^{-i} - 2 \geq \frac{3}{8} M2^{-i}.$$

Finally $s_r \geq 1$ and $1/M = 2^{-(r+1)}$, proving the last assertion. ◀

▶ **Lemma 4.4** (Exact homogeneous output)**.** *Whenever the algorithm reaches its final verified output, that output is a K-clique or a K-vertex independent set.*

Proof. For $i < j$, the definition of S_{i+1} gives $G(v_i, v_j) = c_i$. It also gives $G(v_i, w) = c_i$. Among $r = 2K - 3$ colours, one occurs at least $\lceil r/2 \rceil = K - 1$ times. The corresponding pivots and w therefore induce exactly that colour. The final direct check makes the assertion unconditional on earlier search or estimation errors. ◀

5 Query complexity and the classical landscape

For the size-biased recursion, a capped sample from a nonempty subset of the M-element universe costs $O(\sqrt{M})$ membership tests and $O(\log(2r))$ repetitions give failure probability $O(1/r)$. The predicate at level i costs $O(i+1)$ edge queries, hence one fixed-cap run costs

$$O\left(\sqrt{M}\log(2r)\sum_{i=0}^{r-1}(i+1)\right) = O(K^2 2^K \log K).$$

The repetition in the preceding section proves the following preliminary bound in worst-case, rather than merely expected, query complexity.

▶ **Proposition 5.1** (Size-biased quantum recursion). *Under the hypotheses of Theorem 1.1, the estimation-free algorithm returns a verified homogeneous K-set with probability at least $1-\eta$ using*

$$O\left(K^2 2^K \log K \log \frac{1}{\eta}\right)$$

edge queries.

We next analyze the sharper scale-aware recursion.

By Lemma 4.3, a capped search block at level i uses $O(2^{i/2})$ membership tests. Computing and uncomputing membership and checking the sampled edge costs $O(i+1)$ edge queries. Put

$$b_i = (i+1)2^{i/2}.$$

The batch cap uses $O(m_i + \log(r/\eta)) = O(m_i)$ blocks. Hence its total sampling cost is

$$O\left(\log\frac{r}{\eta}\sum_{i=0}^{r-1} b_i \varepsilon_i^{-2}\right). \tag{5}$$

There are at most six indices at each value of d_i, so

$$Z = \sum_i 2^{-d_i} < 6\sum_{d\geq 0} 2^{-d} = 12.$$

Using Equation (1),

$$\sum_{i=0}^{r-1} b_i\varepsilon_i^{-2} = O\left(\sum_{i=0}^{r-1}(i+1)2^{i/2}4^{d_i}\right).$$

Write $h = r-1-i$. Since $4^{\lceil h/6\rceil} \leq 4\cdot 2^{h/3}$, the last display is bounded by

$$O\left(2^{r/2}\sum_{h=0}^{r-1}(r-h)2^{-h/6}\right) = O(r2^{r/2}).$$

The high-confidence pivot searches, final search, and K^2 verification fit the same bound. Since $r = 2K-3$, this proves Theorem 1.1.

The result counts queries, not total gates or data-loading time. Every search starts from the uniform state on the fixed power-of-two universe $[M]$, not from a pre-prepared state on the unknown S_i. The $O(K)$ measured pivot labels and colours specify the reversible membership tests. No QRAM data structure is assumed, but coherent edge access is: if one edge-oracle evaluation costs C_G gates, its contribution is incurred for each oracle call, not once for the whole algorithm. Constructing such an oracle from an explicit adjacency matrix is outside our query bound.

The known classical lower bound remains below the quantum upper bound. The following direct consequence of the random-Painter analysis of Conlon–Fox–Grinshpun–He [5] records the best bound we obtain from the current literature; details appear in Appendix B.

▶ **Proposition 5.2** (Randomized classical lower bound). *For $K \geq 8$ and $M = 4^{K-1}$, every randomized edge-query algorithm that finds a homogeneous K-set with worst-case probability at least $2/3$ uses*

$$\Omega\left(2^{(2-\sqrt{2})K}\right) = \Omega\left(M^{1-1/\sqrt{2}}\right)$$

queries.

The exponent $1 - 1/\sqrt{2} \approx 0.292893$ does not reach the quantum upper exponent $1/2$, but it strengthens the earlier $M^{1/4}$ black-box lower bound of Impagliazzo–Naor, as recorded in this parameterization by Komargodski–Naor–Yogev [10, 12]. Therefore no quantum-versus-best-randomized-classical separation is claimed. A randomized lower bound $2^{(1+\delta)K}$ for any fixed $\delta > 0$ would, through the online Ramsey connection, improve the long-standing exponential lower bound for diagonal Ramsey numbers. The missing separation is tied to a major combinatorial barrier rather than a routine query lemma.

The size-biased mechanism also extends to any fixed number of edge colours. The statement and its one-line recurrence proof are given in Appendix C.1; this extension is not needed for the sharper binary theorem.

6 Prior art and comparison with a reported lower bound

Previous quantum papers under the phrase “Ramsey algorithm” solve different problems. Gaitan and Clark encode the existence of avoidance colourings in an adiabatic Hamiltonian [8], and Bian et al. implement small instances experimentally [1]; Wang searches the space of $2^{\binom{V}{2}}$ labelled graphs when testing a Ramsey threshold [22]; Qu et al. use quantum counting for hypergraph Ramsey numbers [20]; and generalized/ordinary bound discovery has been studied through adiabatic optimization and QUBO reduction [21, 19]. Their square roots refer to the number of graph colourings, not the number of vertices of one given graph. Dörn’s clique and independent-set query algorithms are closer in oracle model but do not exploit the Ramsey totality promise or the majority recursion [6]. Quantum subset-finding algorithms already give nontrivial clique-query bounds [4], while learning-graph algorithms for fixed subgraph containment [15, 24] and quantum backtracking [17] provide powerful generic primitives.

None by itself supplies the implicit-set survival invariant used here. To our knowledge, no earlier work uses implicit-set quantum sampling to implement the constructive Erdős–Szekeres proof; this priority statement is limited to the exact oracle model and sources compared here.

Krajíček connected Ramsey and weak-pigeonhole proof complexity [13] and later formulated the structured-pigeonhole search problem for circuit-encoded homogeneous sets [14]. Pasarkar, Papadimitriou, and Yannakakis place Ramsey search in their polynomial long-choice class PLC, which captures iterated pigeonhole arguments [18].

Comparison at the formal default parameters.

The closest exact-model source is the FOCS 2024 proceedings paper of Jain–Li–Robere–Xun [11]. Definition II.6 of its proceedings version (p. 413) uses $N = 2^n$ vertices, and the convention after Definition II.7 sets the default target to $K = n/2$. We use these formal definitions throughout this comparison. The unnumbered "Query complexity of RAMSEY" paragraph in Section III (p. 415) states an $N^{1-o(1)}$ quantum lower bound via Lemma III.1 and the Liu–Zhandry multicollision theorem [16]. The graph-hash product in Lemma III.1 can be written off the diagonal as $A_h(u, v) = \mathbf{1}[h(u) = h(v)] \vee A_0(h(u), h(v))$, with $A_h(u, u) = 0$. It is a valid symmetric graph and has an $O(1)$-query coherent implementation from the value oracle for h, using computation and uncomputation of its two values. Thus the promise of a valid graph does not exclude this reduction. Corollary 2.1 also aligns our upper bound with the locally verifiable arbitrary-circuit relation. Moreover, Corollary 1.2 returns $\lfloor n/2 \rfloor + 1$ vertices, so discarding one if necessary solves their default target up to the suppressed rounding convention. Thus malformed encodings and the target convention do not remove the apparent conflict.

Direct parameter substitution.

Write H for the multicollision range size and G for the number of vertices in the Ramsey instance. For fixed multiplicity t, the sufficient parameter condition in Jain et al.'s Theorem I.3 is $G \geq H^{4t}/4^t$. At the boundary $G = \Theta_t(H^{4t})$, Liu–Zhandry's exponent in H is $\alpha_t = (2^{t-1} - 1)/(2^t - 1)$. The inherited exponent is therefore

$$\frac{\alpha_t}{4t} = \frac{2^{t-1} - 1}{4t(2^t - 1)} \leq \frac{1}{24}, \qquad t \geq 2,$$

with equality at $t = 2$. Indeed, $t = 2$ gives $1/24$, while for $t \geq 3$ the strict inequality $\alpha_t < 1/2$ gives $\alpha_t/(4t) < 1/(8t) \leq 1/24$. Increasing G relative to H cannot improve this transferred exponent. The cited Liu–Zhandry result is for fixed t and does not by itself permit substitution of a growing multiplicity. This calculation concerns what follows from these stated ingredients; it is not an upper bound on the true quantum query complexity.

Consequently, the stated near-linear consequence is not obtained by directly composing the two cited statements. If that consequence is intended for the same bounded-error quantum-query relation, it is incompatible with our upper bound. Our proof does not use this consequence or the Liu–Zhandry lower bound. This observation concerns only the parameter substitution in the unnumbered query-complexity discussion and does not concern the principal black-box nonreducibility or TFNP separation theorems of Jain et al. An additional ingredient or a different intended normalization may resolve the discrepancy.

7 Evidence boundary, open problems, and conclusion

The quantum query bound follows from the analytic argument in Sections 2–5; it does not depend on computational verification.

Small proof-of-concept experiments.

Appendix C.2 reports 5,000 ideal-sampler trials at $K = 3, N = 16$, a separate 16-state Grover experiment, and exact finite split-tree comparisons. It describes the experimental settings and outcomes directly. These illustrate the recursion and sampler symmetry rather than establishing a wall-clock or hardware speedup.

Use of AI tools.

GPT-based tools (ChatGPT and Codex) supported brainstorming and constructive discussion of the theoretical arguments and numerical experiments, as well as manuscript polishing and review.

Two algorithmic questions remain. First, determine whether a randomized classical algorithm can match the $\widetilde{O}(\sqrt{M})$ upper bound; Proposition 5.2 leaves a large exponent gap. Second, determine whether the implicit-set method can be adapted to off-diagonal or hypergraph Ramsey total search, and whether the multicolour bound in Corollary C.1 can be sharpened by a scale-aware estimator.

For the hypergraph question, quantum walks already give sublinear-query algorithms for fixed sub-hypergraph containment [9]. As a separate, speculative pointer, higher-order symmetry representations have been useful in a different hypergraph learning setting [23]. Whether such structure can be exposed with sublinear coherent queries in a Ramsey total-search problem is open.

In summary, the constructive Ramsey recursion can be implemented without materializing its exponentially shrinking candidate sets. Capped uniform search and a scale-aware error schedule give a $\widetilde{O}(\sqrt{N})$ query algorithm at the succinct Ramsey scale. This does not imply a new numerical Ramsey bound, a physical speedup, or a quantum-versus-all-classical separation.

A Capped search and batch uniformity

We spell out the exact properties of the search primitive used in Lemma 2.2. Start from the uniform superposition on $[M]$ and use the randomized-iteration unknown-solution search of Boyer–Brassard–Høyer–Tapp [2]. If the marked density is at least λ, truncate the geometric iteration schedule once its maximum iteration count is a sufficiently large constant multiple of $1/\sqrt{\lambda}$. The standard analysis gives a success probability $p_0 > 0$ independent of M and λ, while the total number of membership calls is $O(1/\sqrt{\lambda})$. The truncation is fixed before the oracle outcomes are seen, so the block also terminates when there are no marked states.

For any fixed Grover iteration count, the amplitudes of all marked basis states are equal. Mixing over the iteration count preserves this permutation symmetry. Measure the candidate and evaluate the exact membership predicate; the block succeeds only when this verification accepts. Conditional on acceptance, every marked identity therefore has probability $1/|T|$. A new block reinitializes its registers and random choices, so accepted identities from distinct blocks are independent conditional on the block-success indicators.

For the batch statement, run

$$B = C\bigl(m + \log(1/\delta)\bigr)$$

independent capped blocks, with C an absolute constant chosen using p_0. The number of accepted blocks stochastically dominates $\mathrm{Bin}(B, p_0)$. A Chernoff lower-tail bound makes the probability of fewer than m acceptances at most δ. Conditional on any success pattern with at least m acceptances, the first m accepted identities are independent and uniform on T. This proves Lemma 2.2.

B Derivation of the randomized classical lower bound

We derive Proposition 5.2 from the random-Painter weight bound of Conlon–Fox–Grinshpun–He [5]. Let

$$\beta = 2 - \sqrt{2}, \qquad \alpha = 1 - \frac{1}{\sqrt{2}},$$

so that $\beta = 2\alpha$. Set

$$T = \left\lfloor 2^{\beta K - 2} \right\rfloor, \qquad c = \lfloor \alpha K \rfloor = \alpha K + \epsilon, \qquad -1 < \epsilon \le 0.$$

Expose $G(M, 1/2)$ adaptively to a deterministic T-query algorithm. The cited random-Painter estimate bounds the probability that the queried edges already contain a fully exposed monochromatic K-set by

$$2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}. \tag{6}$$

Because $2T \le 2^{\beta K - 1}$, the base-2 logarithm of Equation (6) is at most

$$1 - \binom{K}{2} + c(c-1) + (K-c)(\beta K - 1) = 1 - \frac{K}{2} + \epsilon^2.$$

For $K \geq 8$ this is at most -2, so the event has probability at most $1/4$.

If the algorithm outputs a K-set that is not already a fully exposed monochromatic set, condition on the transcript. If the exposed internal edges contain both colours, the set cannot be homogeneous. If they contain exactly one colour, at least one unexposed fair edge must match that colour, so the conditional success probability is at most $1/2$. If no internal edge has been exposed, the probability is $2^{1-\binom{K}{2}} \leq 1/2$ for $K \geq 3$. Thus the conditional probability of a homogeneous output is at most $1/2$ in every case relevant here. The total success probability under $G(M, 1/2)$ is at most

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8} < \frac{2}{3}.$$

Yao's minimax principle now implies that any randomized algorithm with worst-case success probability at least $2/3$ needs more than T queries. Since $M = 4^{K-1}$,

$$2^{\beta K} = \Theta\left(M^{\beta/2}\right) = \Theta\left(M^{1-1/\sqrt{2}}\right),$$

which proves Proposition 5.2.

C Multicolour extension and numerical experiments

C.1 A multicolour extension of size-biased survival

The size-biased mechanism is not specific to two colours. Suppose every edge has one of $q \geq 2$ colours, and let

$$r = q(K-2) + 1, \qquad B_d = 1 + q + \cdots + q^{d-1} = \frac{q^d - 1}{q - 1}.$$

Choose the power of two

$$L_q = 2^{\lceil \log_2(2B_r) \rceil},$$

so $2B_r \leq L_q < 4B_r$. If the $s-1$ non-pivot vertices split into sizes $a_1, \ldots, a_q$, the induction in Lemma 4.1 becomes

$$\sum_{j=1}^{q} (a_j - B_{d-1})_+ \geq (s - 1 - qB_{d-1})_+ = (s - B_d)_+,$$

because $B_d = 1 + qB_{d-1}$. Thus, for $d \geq 1$, the d-step survival probability is at least $(s - B_d)_+/(s-1)$ when $s \geq 2$. Starting from $L_q \geq 2B_r$ gives probability greater than $1/2$, and among the r recorded colours one occurs at least $K-1$ times.

▶ **Corollary C.1** (Multicolour size-biased Ramsey search)**.** *For $q, K \geq 2$, every q-edge-coloured complete graph on at least L_q vertices admits a bounded-error quantum algorithm that outputs a monochromatic K-clique using*

$$O\left(r^2 \sqrt{L_q} \log(2r) \log \frac{1}{\eta}\right)$$

colour-oracle queries in the worst case, where r, B_r, and L_q are as defined above.

C.2 Small-instance numerical experiments

Ideal-sampler recursion. We ran the scale-aware recursion at $K = 3$, $N = 16$, and $r = 3$ for 1,000 trials in each of five graph families. The error schedule is $(\varepsilon_0, \varepsilon_1, \varepsilon_2) = (1/64, 1/64, 1/32)$, with sample counts $m_i = \lceil \log(600)/(2\varepsilon_i^2) \rceil$; this assigns total estimation-failure budget 0.01. Pivots are uniform, majority-estimation samples are independent with replacement, and ties select colour 1. Quantum search is replaced by ideal uniform sampling, so this tests the combinatorial recursion, not coherent search failures or query costs.

The deterministic inputs are the complete graph, the empty graph, and the balanced complete bipartite graph $K_{8,8}$. For the random inputs, one $G(16, 1/2)$ graph is held fixed across all 1,000 trials, whereas the final family draws a fresh $G(16, 1/2)$ graph for each trial. Every returned triple is checked; exhaustion or a nonhomogeneous output counts as failure.

Graph family on 16 vertices	failed/trials	minimum final $\lvert S_3 \rvert$
complete	0/1000	13
empty	0/1000	13
balanced complete bipartite $K_{8,8}$	0/1000	6
one fixed $G(16, 1/2)$	0/1000	2
fresh $G(16, 1/2)$ per trial	0/1000	2

All 5,000 trials returned homogeneous triples. The table also reports the smallest final candidate-set size observed within each family. Zero observed failures do not establish zero failure probability.

Isolated Grover sampler. We separately evolved a 16-dimensional state vector initialized to the uniform superposition, with marked-set sizes $1, 2, 3, 4, 8, 15$. We averaged the measurement distributions over a uniformly chosen number of Grover iterations from zero through four. The probability of measuring a marked item ranged from 0.4026746749 to 0.5973253251. Conditional on that outcome, the largest deviation from the uniform distribution on marked items was zero to machine precision. This experiment illustrates the symmetry used in Lemma 2.2; it does not simulate its full capped search schedule.

Finite split-tree survival. Minimizing the ideal size-biased survival probability over every admissible colour-class split gives exact finite-depth comparisons with our bounds. For nine binary levels starting with 1,024 candidates, the minimum was approximately 0.5086135566, above the proved lower bound 513/1024. For four ternary levels starting with 128 candidates, it was approximately 0.7155732457, above 88/127. These are small-instance checks, not full coherent simulations, hardware experiments, or speedup measurements.

References

1 Zhengbing Bian, Fabian Chudak, William G. Macready, Geordie Rose, and Frank Gaitan. Experimental determination of Ramsey numbers. *Physical Review*

Letters, 111:130505, 2013. URL: `https://arxiv.org/abs/1201.1842`, `arXiv:1201.1842`, `doi:10.1103/PhysRevLett.111.130505`.

2 Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp. Tight bounds on quantum searching. *Fortschritte der Physik*, 46(4–5):493–506, 1998. URL: `https://arxiv.org/abs/quant-ph/9605034`, `arXiv:quant-ph/9605034`.

3 Marcelo Campos, Simon Griffiths, Robert Morris, and Julian Sahasrabudhe. An exponential improvement for diagonal Ramsey. *Annals of Mathematics*, 203(3):869–932, 2026. `arXiv:2303.09521`, `doi:10.4007/annals.2026.203.3.4`.

4 Andrew M. Childs and Jason M. Eisenberg. Quantum algorithms for subset finding. *Quantum Information and Computation*, 5(7):593–604, 2005. `arXiv:quant-ph/0311038`, `doi:10.26421/QIC5.7-7`.

5 David Conlon, Jacob Fox, Andrey Grinshpun, and Xiaoyu He. Online Ramsey numbers and the subgraph query problem. In *Building Bridges II: Mathematics of László Lovász*, volume 28 of *Bolyai Society Mathematical Studies*, pages 159–194. Springer, 2020. URL: `https://arxiv.org/abs/1806.09726`, `arXiv:1806.09726`.

6 Sebastian Dörn. Quantum complexity bounds for independent set problems. *arXiv preprint*, 2005. URL: `https://arxiv.org/abs/quant-ph/0510084`, `arXiv:quant-ph/0510084`.

7 Paul Erdős and George Szekeres. A combinatorial problem in geometry. *Compositio Mathematica*, 2:463–470, 1935. URL: `https://eudml.org/doc/88611`.

8 Frank Gaitan and Lane Clark. Ramsey numbers and adiabatic quantum computing. *Physical Review Letters*, 108:010501, 2012. URL: `https://arxiv.org/abs/1103.1345`, `arXiv:1103.1345`.

9 François Le Gall, Harumichi Nishimura, and Seiichiro Tani. Quantum algorithms for finding constant-sized sub-hypergraphs. *Theoretical Computer Science*, 609:569–582, 2016. `arXiv:1310.4127`, `doi:10.1016/j.tcs.2015.10.006`.

10 Russell Impagliazzo and Moni Naor. Decision trees and downward closures. In *Proceedings of the Third Annual Structure in Complexity Theory Conference*, pages 29–38. IEEE, 1988. `doi:10.1109/SCT.1988.5260`.

11 Siddhartha Jain, Jiawei Li, Robert Robere, and Zhiyang Xun. On pigeonhole principles and Ramsey in TFNP. In *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 406–428. IEEE, 2024. `arXiv:2401.12604`, `doi:10.1109/FOCS61266.2024.00033`.

12 Ilan Komargodski, Moni Naor, and Eylon Yogev. White-box vs. black-box complexity of search problems: Ramsey and graph property testing. *Journal of the ACM*, 66(5):34:1–34:28, 2019. `doi:10.1145/3341106`.

13 Jan Krajíček. On the weak pigeonhole principle. *Fundamenta Mathematicae*, 170(1–2):123–140, 2001. `doi:10.4064/fm170-1-8`.

14 Jan Krajíček. Structured pigeonhole principle, search problems and hard tautologies. *The Journal of Symbolic Logic*, 70(2):619–630, 2005. `doi:10.2178/jsl/1120224731`.

15 Troy Lee, Frédéric Magniez, and Miklos Santha. A learning graph based quantum query algorithm for finding constant-size subgraphs. *Chicago Journal of Theoretical Computer Science*, 2012(10), 2012. URL: `https://arxiv.org/abs/1109.5135`, `arXiv:1109.5135`.

16 Qipeng Liu and Mark Zhandry. On finding quantum multi-collisions. In *Advances in Cryptology—EUROCRYPT 2019, Part III*, volume 11478 of *Lecture Notes in Computer Science*, pages 189–218. Springer, 2019. `arXiv:1811.05385`, `doi:10.1007/978-3-030-17659-4_7`.

17 Ashley Montanaro. Quantum walk speedup of backtracking algorithms. *Theory of Computing*, 14(15):1–24, 2018. URL: `https://arxiv.org/abs/1509.02374`, `arXiv:1509.02374`, `doi:10.4086/toc.2018.v014a015`.

18 Amol Pasarkar, Christos Papadimitriou, and Mihalis Yannakakis. Extremal combinatorics, iterated pigeonhole arguments and generalizations of PPP. In *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*, volume 251 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 88:1–88:20. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2023. `doi:10.4230/LIPIcs.ITCS.2023.88`.

19 Joel E. Pion and Susan M. Mniszewski. Toward computing bounds for Ramsey numbers using quantum annealing. *Quantum Information Processing*, 24(7):221, 2025. URL: `https://arxiv.org/abs/2311.04405`, `arXiv:2311.04405`, `doi:10.1007/s11128-025-04839-x`.

20 Ri Qu, Zong shang Li, Juan Wang, Yan ru Bao, and Xiao chun Cao. Computing hypergraph Ramsey numbers by using quantum circuit. *Quantum Information Processing*, 12(7):2487–2496, 2013. `arXiv:1210.3419`, `doi:10.1007/s11128-013-0541-9`.

21 Mani Ranjbar, William G. Macready, Lane Clark, and Frank Gaitan. Generalized Ramsey numbers through adiabatic quantum optimization. *Quantum Information Processing*, 15(9):3519–3542, 2016. `arXiv:1606.01078`, `doi:10.1007/s11128-016-1363-3`.

22 Hefeng Wang. Determining Ramsey numbers on a quantum computer. *Physical Review A*, 93(3):032301, 2016. `arXiv:1510.01884`, `doi:10.1103/PhysRevA.93.032301`.

23 Simon Zhang, Cheng Xin, and Tamal K. Dey. Expressive higher-order link prediction through hypergraph symmetry breaking. *Transactions on Machine Learning Research*, November 2024. URL: `https://openreview.net/forum?id=oG65SjZNIF`.

24 Yechao Zhu. Quantum query complexity of constant-sized subgraph containment. *International Journal of Quantum Information*, 10(3):1250019, 2012. `arXiv:1109.4165`, `doi:10.1142/S0219749912500190`.

AI Disclosure. We used GPT-based tools for brainstorming, constructive discussion, manuscript polishing, and review. The author takes full responsibility for the content.